

# A Level Computer Science (9618)

Topical Past Paper Worksheet

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## Topic 4: Processor Fundamentals

Subtopic 4.1: Central Processing Unit (CPU) Architecture

**Concept 4.1.4: System Buses (Address, Data, Control)**

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Total Questions: 3

**Mark Schemes begin on Page 4**

Question 1 | Source: 9618\_s25\_qp\_11 (Q6.a.iii)

6 A programmer is buying a new computer.

(a) The programmer is considering the following two computers:

|            |                                                                                                                    |
|------------|--------------------------------------------------------------------------------------------------------------------|
| Computer 1 | Quad-core 2.2 GHz processor<br>16-bit architecture<br>1 GB Random Access Memory (RAM)<br>500 GB magnetic hard disk |
| Computer 2 | Dual-core 3.8 GHz processor<br>32-bit architecture<br>2 GB RAM<br>500 GB solid state drive (SSD)                   |

(iii) The two computers have different bus widths.

Explain how different bus widths affect the performance of a computer.

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.....

..... [2]

Question 2 | Source: 9618\_s23\_qp\_12 (Q5.a)

5 A student has purchased a new laptop.

(a) The laptop is designed using the Von Neumann model for a computer system.

Identify **two** types of signal that a control bus can transfer.

1 .....

2 ..... [2]

Question 3 | Source: 9618\_w22\_qp\_11 (Q5.b.ii)

(b) A Central Processing Unit (CPU) contains several special purpose registers and other components.

(ii) Tick (✓) **one** box in each row to identify the system bus used by each CPU component.

| <b>CPU component</b>          | <b>Data bus</b> | <b>Address bus</b> | <b>Control bus</b> |
|-------------------------------|-----------------|--------------------|--------------------|
| System clock                  |                 |                    |                    |
| Memory Address Register (MAR) |                 |                    |                    |

[1]

# MARK SCHEMES

## Mark Scheme for Question 1 | Source: 9618\_s25\_ms\_11 (Q6.a.iii)

|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 6(a)(iii) | <b>1 mark</b> each to <b>max 2</b><br><br>e.g. <ul style="list-style-type: none"> <li>• Wider data bus means more data can be transferred between components at a time</li> <li>• there is less delay / latency when fetching data for a running process</li> <li>• Wider address bus means larger memory addresses can be used</li> <li>• allowing more memory locations to be accessed directly</li> <li>• so less likely to run out of memory</li> </ul> | <b>2</b> |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|

## Mark Scheme for Question 2 | Source: 9618\_s23\_ms\_12 (Q5.a)

|      |                                                                                                                                                                      |          |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 5(a) | <b>1 mark</b> each to <b>max 2</b><br><br>Examples: <ul style="list-style-type: none"> <li>• Interrupt</li> <li>• Timing</li> <li>• Read</li> <li>• Write</li> </ul> | <b>2</b> |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|

## Mark Scheme for Question 3 | Source: 9618\_w22\_ms\_11 (Q5.b.ii)

5(b)(ii)

1 mark for both rows:

| CPU component                 | Data bus | Address bus | Control bus |
|-------------------------------|----------|-------------|-------------|
| System clock                  |          |             | ✓           |
| Memory Address Register (MAR) |          | ✓           |             |

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